Knowing when to ask a human: reliable confidence for fast time series classifiers under sensor faults

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See separate title page.

Abstract

Monitoring systems, from heart monitors to industrial sensors, often let a time series classifier decide on its own and pass uncertain cases to a person. This only works if the classifier's confidence drops when a recording is damaged by noise, drift, spikes or dropouts. Fast random convolutional kernel classifiers such as MiniRocket are accurate and cheap, but their ridge readout gives scores rather than probabilities, and existing calibration methods rescale all scores in the same way. We propose LOOM, a closed-form readout that divides each ridge score by the square root of its leverage, a measure of how far a recording lies from the training data, and fits one temperature and the ridge penalty on exact leave-one-out pairs of scores and leverages, without held-out data, refitting or ensembles. We tested it on 114 UCR datasets with 18 simulated sensor faults, on seven ECG datasets with real recorded noise from the MIT-BIH Noise Stress Test Database, and in a pre-registered triage study on 38 sensor datasets. Under simulated faults LOOM lowered the median negative log-likelihood from 1.728 to 0.738 compared with prevalidated ridge (PreVal), the strongest previous readout; under real ECG noise from 3.346 to 0.678; and with a 20 % review budget its review queue caught 41 % of the faulty recordings, against 24 % for PreVal and 34 % for the tree-based QUANT classifier. On clean data LOOM is not better than PreVal, and QUANT has a lower likelihood loss under the strongest faults.

Keywords: Time series classification, Sensor faults, Uncertainty estimation, Probability calibration, Human-in-the-loop triage, Conformal prediction

1 Introduction

Consider a wearable heart monitor that labels each short ECG recording as normal or abnormal, a factory system that classifies the signal of a production sensor, or a watch that recognises gestures from an accelerometer. Such systems make many decisions

per day and nobody can check each one. A common safeguard is to let the classifier decide alone when it is sure and to pass the other cases to a person. This works only if the confidence of the classifier means something: high when the prediction is likely to be right and low when it is not (Abdar et al. 2021; Gawlikowski et al. 2023).

Real sensor data make this hard. Electrodes come loose, cables pick up interference, a patient moves, a sensor drifts or drops samples. A recording damaged in this way can look unlike any training example, yet most classifiers still return a confident answer. For a triage system, a confident wrong answer is the worst case, because it never reaches the reviewer.

We study this problem for random convolutional kernel classifiers. ROCKET (Dempster et al. 2020) and its successors MiniRocket (Dempster et al. 2021), MultiRocket (Tan et al. 2022) and Hydra (Dempster et al. 2023) map each series to thousands of features from random convolutions and fit a ridge regression on top. They are about as accurate as much larger ensembles and deep networks on the UCR archive (Dau et al. 2019; Ismail Fawaz et al. 2019; Middlehurst et al. 2024b) and a model trains in a median of 0.31 s on one CPU core in our experiments (Sect. 6.11), which makes them attractive for monitoring on modest hardware. Their ridge readout, however, gives scores rather than probabilities. Platt scaling, isotonic regression and temperature scaling (Guo et al. 2017; Platt 2000; Zadrozny and Elkan 2002) need held-out data; the Arsenal (Middlehurst et al. 2021) needs an ensemble; and prevalidated ridge regression (PreVal) (Dempster et al. 2025) rescales all scores by one factor fitted on exact leave-one-out (LOO) predictions. All of them treat a damaged recording like a clean one.

Ridge regression already measures how far a new input lies from its training data: the *leverage*, a classical diagnostic of extrapolation (Hoaglin and Welsch 1978) and the predictive variance factor of the equivalent Bayesian linear model (MacKay 1992). We show that the algebra behind the LOO shortcut also gives the *held-out* leverage of every training example, so that a leverage-aware calibration can be fitted on exact out-of-sample pairs without any held-out data. The resulting readout, LOOM (leave-one-out fitted, leverage-moderated), divides each ridge score by the square root of its relative leverage.

1.1 Motivation and contributions

The gap we address is that no readout for fast random-kernel classifiers lowers its confidence specifically for recordings that lie far from the training data, and none has been tested for fault-aware triage. Our contributions are:

1. LOOM, a closed-form probabilistic readout whose confidence drops for recordings far from the training data, fitted on exact LOO scores and held-out leverages from one singular value decomposition (SVD) (Sect. 4).
2. Exact jackknife+ prediction sets for ridge readouts, computed without refitting (Sect. 4.5).
3. A pre-registered evaluation with explicit hypotheses: 114 datasets and 18 simulated sensor faults, real recorded ECG noise, readouts on non-convolutional features, ablations, sanity checks, and frequentist and Bayesian statistics (Sects. 5 and 6).

4. A triage case study with an exploratory phase on 12 datasets and a pre-registered confirmatory replication on 38 further datasets (Sect. 6.5).
5. An open-source command-line tool that turns labelled recordings into a triage system with a review queue (Sect. 6.12).

The ingredients—LOO identities, leverage and moderated outputs—are classical; our contribution is to combine them into a readout that needs no held-out data and to test it for triage under sensor faults. We also report where LOOM does not help (Sect. 6.13).

1.2 Organization

Section 2 reviews related work, Sect. 3 introduces notation and assumptions, Sect. 4 describes LOOM, Sect. 5 the experimental framework and Sect. 6 the results. Section 7 concludes. Proofs, the deviations log and further results are in the Supplementary Material (Sects. S1–S6).

2 Literature review

2.1 Random convolutional kernel classifiers

ROCKET (Dempster et al. 2020) applies 10,000 random dilated kernels and keeps the maximum and the proportion of positive values of each output. MiniRocket (Dempster et al. 2021) makes the transform almost deterministic and much faster, MultiRocket (Tan et al. 2022) adds pooling operators and the first-difference series, and Hydra (Dempster et al. 2023) lets groups of kernels compete. All use a linear readout. The bake-off of Middlehurst et al. (2024b) confirms their accuracy–cost trade-off and reports log-loss, but not calibration under shift. Probabilistic alternatives include the Arsenal (Middlehurst et al. 2021), QUANT (Dempster et al. 2024) and deep networks such as InceptionTime (Ismail Fawaz et al. 2020).

2.2 Calibration and prevalidation

Platt scaling (Platt 2000), isotonic regression (Niculescu-Mizil and Caruana 2005; Zadrozny and Elkan 2002), temperature scaling (Guo et al. 2017) and Dirichlet calibration (Kull et al. 2019) map scores to probabilities with held-out or cross-validated scores (Silva Filho et al. 2023). Fitting a second model to cross-validated predictions is known as pre-validation (Tibshirani and Efron 2002). For least squares, LOO predictions have a closed form (Allen 1974; Cawley and Talbot 2004; Golub et al. 1979). PreVal (Dempster et al. 2025) uses them to fit a global scale factor for ridge regression; it is the work closest to ours.

2.3 Uncertainty when the data change

Globally calibrated models become overconfident under dataset shift, while ensembles hold up better (Lakshminarayanan et al. 2017; Ovadia et al. 2019). Distance-aware networks such as SNGP (Liu et al. 2020, 2023) and DUQ (van Amersfoort et al. 2020)

and last-layer Laplace approximations (Kristiadi et al. 2020) become uncertain away from the data, following the moderated outputs of MacKay (1992), but require changes to a network's training. Corruption benchmarks (Hendrycks and Dietterich 2019) and confidence-based detection of unusual inputs (Hendrycks and Gimpel 2017) are standard for images; for time series, perturbations are mostly used for augmentation (Iwana and Uchida 2021; Um et al. 2017) and domain adaptation assumes unlabelled target data (Ragab et al. 2023). Selective classification studies the trade-off between coverage and risk (Geifman and El-Yaniv 2017).

2.4 Conformal prediction

Conformal prediction gives prediction sets with a coverage guarantee under exchangeability (Angelopoulos and Bates 2023; Vovk et al. 2022). Split conformal prediction sets data aside (Lei et al. 2018); jackknife+ uses all LOO models (Barber et al. 2021). APS (Angelopoulos et al. 2021; Romano et al. 2020) and LAC (Sadinle et al. 2018) are common scores. For ridge regression, conformal prediction has a closed form (Burnaev and Vovk 2014; Nouretdinov et al. 2001); time series work has focused on forecasting (Xu and Xie 2023).

2.5 Motivations

Table 1 summarises the properties of the closest methods. Five motivations follow: (i) the fast random-kernel classifiers lack a probabilistic readout that responds to unusual inputs; (ii) held-out calibration data are scarce in small monitoring datasets; (iii) ensembles and distance-aware networks are costly for embedded use; (iv) the closed-form statistics needed for an input-dependent readout already exist but have not been combined for this purpose; and (v) no study has evaluated readouts for fault-aware triage with real sensor noise.

3 Preliminary concepts

We observe n training series $x_i \in \mathbb{R}^L$ with labels $y_i \in \{1, \dots, K\}$. A random-kernel transform $\varphi : \mathbb{R}^L \to \mathbb{R}^p$ ($p \approx 10^4$) is fitted on the training series. Features are standardised with the training mean and standard deviation; $\psi(x)$ denotes the centred standardised features, $\tilde{x}(x) = (1, \psi(x))$ the augmented vector, $\tilde{X}$ the matrix of training vectors, Φ_c the centred training matrix and $Y \in \{-1, +1\}^{n \times K}$ the one-vs-rest targets. Ridge regression (Hoerl and Kennard 1970) with an unpenalised intercept and penalty λ gives

$$A = \tilde{X}^\top \tilde{X} + \operatorname{diag}(0, \lambda I_p), \qquad B = A^{-1} \tilde{X}^\top Y, \qquad \hat{Y} = \tilde{X} B = HY. \tag{1}$$

With the thin SVD $\Phi_c = USV^\top$ (rank $r \le \min(n-1, p)$) and $D_\lambda = \operatorname{diag}\big(1/(s_j^2 + \lambda)\big)$, the hat diagonal d_i, the scores $z(x) = B^\top \tilde{x}(x)$ and the leverage $h(x) = \tilde{x}(x)^\top A^{-1} \tilde{x}(x)$

Table 1 Properties of the closest methods. ✓: yes; (✓): partly; –: no. "Fault-aware" means that confidence decreases for inputs far from the training data; "tested on faults" means an evaluation under sensor faults or corruptions of time series.

Method	Proba-bilities	No held-out data	No refit/ ensemble	Fault-aware	Prediction sets	Tested on faults
Ridge readout (Dempster et al. 2021)	–	✓	✓	–	–	–
Platt / isotonic / temperature (Guo et al. 2017; Platt 2000; Zadrozny and Elkan 2002)	✓	–	(✓)	–	–	–
PreVal (Dempster et al. 2025)	✓	✓	✓	–	–	–
Arsenal (Middlehurst et al. 2021)	✓	✓	–	(✓)	–	–
QUANT (Dempster et al. 2024)	✓	✓	–	–	–	–
Deep ensembles (Lakshminarayanan et al. 2017)	✓	✓	–	(✓)	–	(✓)
SNGP / DUQ / Laplace (Kristiadi et al. 2020; Liu et al. 2020; van Amersfoort et al. 2020)	✓	✓	–	✓	–	–
Jackknife+ (Barber et al. 2021)	–	✓	–	–	✓	–
LOOM (ours)	✓	✓	✓	✓	✓	✓

are

$$d_i = \frac{1}{n} + \sum_{j=1}^{r} u_{ij}^2 \frac{s_j^2}{s_j^2 + \lambda}, \qquad h(x) = \frac{1}{n} + \left\| D_\lambda^{1/2} V^\top \psi \right\|^2 + \frac{\|\psi\|^2 - \|V^\top \psi\|^2}{\lambda}. \tag{2}$$

The last term measures the part of $\psi(x)$ outside the span of the training data; it dominates when $p \gg n$.

We make the following assumptions.

Assumption A1 *Clean training and clean test recordings are exchangeable.*

Assumption A2 *Under Assumption A1, the held-out leverages h_{-i} of the training examples are distributed like the leverages $h(x)$ of clean test recordings. (Checked empirically in Sect. 6.10.)*

Assumption A3 *Sensor faults move the features of a recording away from the span of the training features, so that its leverage grows. (Holds for some fault types but not all; Sects. 6.3 and 6.10.)*

Assumption A4 *The standardisation and the kernel biases are treated as fixed when LOO quantities are computed. (They are estimated on all n series, so the LOO identities are exact only conditionally on them.)*

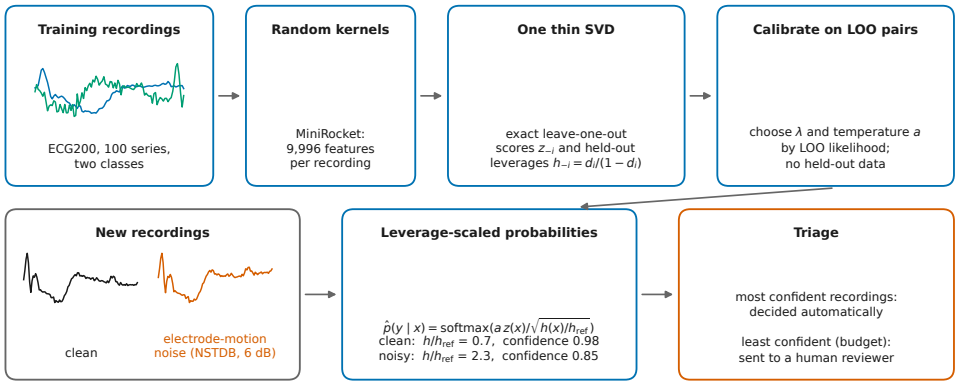


Fig. 1 LOOM turns the ridge readout of a random-kernel classifier into probabilities whose confidence drops for unusual recordings. Examples are real: the first training series of each class of ECG200 and the first test series in file order, shown clean and with real electrode-motion noise from the MIT-BIH Noise Stress Test Database at 6 dB; the printed leverage ratios and confidences are computed by the fitted model.

Algorithm 1 LOOM: fitting and triage

Require: training series $\{(x_i, y_i)\}_{i=1}^{n}$, penalty grid Λ, review budget b

Fit the random-kernel transform; compute standardised features $\psi(x_i)$
Thin SVD $\Phi_c = USV^\top$ ▷ one decomposition for all λ
for $\lambda \in \Lambda$ **do**
    d_i by Eq. (2); z_{-i} and h_{-i} by Eq. (3); $h_{\mathrm{ref}} \leftarrow \mathrm{median}(h_{-i})$
    $a_\lambda \leftarrow \arg\min_a \mathcal{L}(a, \lambda)$ by Eq. (5) with smoothed targets
end for
$(\lambda^\star, a^\star) \leftarrow \arg\min_\lambda \mathcal{L}(a_\lambda, \lambda)$
for each new recording x **do**
    $\hat{p}(\cdot \mid x)$ by Eq. (4) with $\lambda^\star, a^\star$; optional jackknife+ set by Eq. (6)
end for
Send the b share of recordings with the lowest $\max_k \hat{p}(k \mid x)$ to review; decide the rest automatically

4 Proposed methodology

4.1 Overview

Figure 1 shows the pipeline and Algorithm 1 the procedure. One thin SVD gives, for every candidate penalty, the exact LOO scores and held-out leverages of all training series; the penalty and a temperature are fitted on these pairs; and at test time each score is divided by the square root of the relative leverage. The resulting confidence decides whether a recording is handled automatically or sent to review.

4.2 Exact leave-one-out scores and held-out leverages

Let B_{-i} denote the fit without example i (same λ and standardisation).

Proposition 1 (LOO scores and held-out leverages)

$$z_{-i} := B_{-i}^{\top}\tilde{x}_i = Y_i - \frac{Y_i - \hat{Y}_i}{1 - d_i}, \qquad h_{-i} := \tilde{x}_i^{\top} A_{-i}^{-1}\tilde{x}_i = \frac{d_i}{1 - d_i}. \tag{3}$$

The first identity is the classical PRESS shortcut (Allen 1974) used by RidgeClassifierCV and PreVal; the second follows from the same Sherman–Morrison update and is the quantity LOOM adds. Proofs are in Supplementary Sect. S1. The pair (z_{-i}, h_{-i}) is an exact out-of-sample observation of a score and its uncertainty, available for all n examples at $O(nr)$ cost.

4.3 Leverage-scaled probabilities

Under the Bayesian linear model with prior $\beta \sim \mathcal{N}(0, \sigma^2\lambda^{-1}I)$, the predictive standard deviation of each score is $\sigma\sqrt{h(x)}$. LOOM divides the scores by this quantity relative to $h_{\mathrm{ref}} = \mathrm{median}(h_{-i})$:

$$\hat{p}(y = k \mid x) = \mathrm{softmax}_k\left(\frac{a\, z_k(x)}{\sqrt{h(x)/h_{\mathrm{ref}}}}\right). \tag{4}$$

For recordings like the training data $h(x) \approx h_{\mathrm{ref}}$ and Eq. (4) is a temperature-scaled ridge; when a fault moves a recording away from the training span, $h(x)$ grows and the probabilities move towards uniform. Eq. (4) is the limit $c \to \infty$ of the moderation $\mathrm{softmax}\big(az/\sqrt{1 + ch/h_{\mathrm{ref}}}\big)$ (Kristiadi et al. 2020; MacKay 1992); the limit was fixed on eight development datasets before the benchmark, and finite c is an ablation.

4.4 Fitting on exact LOO pairs

For each λ the temperature minimises the LOO negative log-likelihood

$$\mathcal{L}(a, \lambda) = -\frac{1}{n}\sum_{i=1}^{n}\sum_{k=1}^{K} t_{ik} \log \mathrm{softmax}_k\left(\frac{a\, z_{-i,k}}{\sqrt{h_{-i}/h_{\mathrm{ref}}}}\right), \tag{5}$$

and the pair (λ, a) with the smallest value is kept. When the LOO predictions separate the classes, the optimal temperature is infinite; we therefore smooth the targets following Platt (2000): the true class receives $(n_k + 1)/(n_k + 2)$, where n_k is its training count, and the rest is spread evenly. The one-dimensional problem in $\log a$ is solved with L-BFGS from three starting values.

4.5 Closed-form jackknife+ prediction sets

With a non-conformity score $s(x, y; f)$ (LAC, $1 - \hat{p}(y \mid x)$ (Sadinle et al. 2018), or deterministic APS (Romano et al. 2020)), jackknife+ returns $C(x) = \{y : \#\{i : s_i <$

$s(x, y; f_{-i})\} < (1-\alpha)(n+1)\}$ with $s_i = s(x_i, y_i; f_{-i})$ (Barber et al. 2021). It needs every LOO model at every test point.

Proposition 2 (LOO predictions at a new point) *With* $k(x,i) = \tilde{x}(x)^\top A^{-1}\tilde{x}_i = 1/n + (V^\top\psi)^\top D_\lambda S u_i$ *and residuals* $E = Y - \hat{Y}$,

$$z_{-i}(x) = z(x) - \frac{k(x,i)E_i}{1-d_i}, \qquad h_{-i}(x) = h(x) + \frac{k(x,i)^2}{1-d_i}. \tag{6}$$

This parallels closed-form conformal ridge regression (Nouretdinov et al. 2001). Because λ, a and the standardisation use all n examples (Assumption A4), the coverage guarantee holds only approximately; we check it empirically (Sect. 6.8).

4.6 Complexity

The SVD costs $O(np\min(n,p))$, as for RidgeClassifierCV; ten penalties and the temperature fits add $O(10\,nrK)$; prediction costs $O(p(r+K))$ per series instead of $O(pK)$; memory is $O((n+p)r)$. Cross-validated calibration needs k further fits and an M-member ensemble M transforms and fits.

5 Experimental framework

5.1 Setup

Code is in Python (NumPy, SciPy, scikit-learn (Pedregosa et al. 2011), aeon 1.6.0 (Middlehurst et al. 2024a)). Every job ran on one core of a 4-core cloud CPU without a GPU (14.3 GB memory limit); times are single-core wall-clock times. One script regenerates all results, tables, figures and the number macros of this paper; unit tests compare Eqs. (2), (3) and (6) with explicit refits. The method was developed on eight datasets and the pipeline tested on six more; results without them are reported. The protocol was registered before the benchmark, and every later change is logged (Supplementary Sect. S2). An AI assistant (Claude, Anthropic) was used for the literature search, for writing and running code and for drafting text; all numbers come from the released result files, and the authors take responsibility for the content.

5.2 Datasets

Table 2 lists the data. The benchmark uses the 114 univariate equal-length UCR datasets (Dau et al. 2019) with the official split and four stratified resamples (three for the four largest datasets), following the bake-off protocol (Bagnall et al. 2016). A leakage audit found exact train/test duplicates in 16 datasets and exact or near duplicates in 24; a planted-leak test confirmed that the audit detects such cases (Supplementary Sect. S3).

5.3 Faults and the triage protocol

Simulated faults are applied to test series only, with amplitudes relative to each series' standard deviation: noise, baseline wander or drift, dropout (a segment replaced by

Table 2 Data used in each experiment. All datasets are from the UCR archive; real noise is from the MIT-BIH Noise Stress Test Database.

Experiment	Datasets	Splits	Purpose
Benchmark (E1–E9)	114 univariate, equal length	official + 4 resamples	calibration, faults, ablation, prediction sets
Real ECG noise (E12)	7 ECG datasets	official + 4 resamples	real recorded noise, 9 conditions
Triage, exploratory (E11)	12 sensor datasets, 3 domains	official + 4 resamples	review-queue study
Triage, confirmatory (E13)	38 further sensor datasets	official + 4 resamples	pre-registered replication
QUANT features (E14)	110 datasets ($\leq$ 3,000 training series)	official split	non-convolutional features
Natural change (E10)	UWaveGestureLibrary X/Y/Z	official + 4 resamples	cross-axis sensor change

interpolation), spikes, speed variation (smooth time warp) and misalignment (circular shift), each at three severities (Supplementary Sect. S4). *Real noise*: baseline wander, electrode motion and muscle artefact recordings of the MIT-BIH Noise Stress Test Database (Goldberger et al. 2000; Moody et al. 1984) are added at 12, 6 and 0 dB SNR to the seven ECG datasets. *Triage*: each test recording is faulty with probability 0.3, with a domain-specific fault at a random severity; the share b of recordings with the lowest confidence goes to review.

5.4 Metrics

With predicted probabilities $\hat{p}_{ik}$ and true labels y_i on m test recordings, we report the negative log-likelihood, the expected calibration error over 15 equal-width bins $B_1, \dots, B_{15}$ of the top-label confidence, and the Brier score:

$$\mathrm{NLL} = -\frac{1}{m}\sum_{i} \log \hat{p}_{iy_i}, \qquad \mathrm{Brier} = \frac{1}{m}\sum_{i,k} \big(\hat{p}_{ik} - \mathbb{1}[y_i = k]\big)^2, \tag{7}$$

$$\mathrm{ECE} = \sum_{j} \frac{|B_j|}{m} \big|\mathrm{acc}(B_j) - \mathrm{conf}(B_j)\big|. \tag{8}$$

Fault detection is measured by the AUROC of $1 - \max_k \hat{p}_{ik}$ for separating faulty from clean recordings. For prediction sets we report coverage $\frac{1}{m}\sum_i \mathbb{1}[y_i \in C(x_i)]$ and mean set size. For triage with review set R ($|R| = bm$) and faulty set F:

$$\text{faults caught} = \frac{|R \cap F|}{|F|}, \qquad \text{automatic error} = \frac{|\{i \notin R : \hat{y}_i \neq y_i\}|}{m - |R|}. \tag{9}$$

5.5 Baselines

All readouts use identical MiniRocket features: ridge one-hot output, softmax of the ridge scores, PreVal (Dempster et al. 2025) (reimplemented without smoothing, as published), hold-out temperature scaling (80/20), five-fold Platt and isotonic calibration (Pedregosa et al. 2011), cross-validated logistic regression, and five-seed ensembles

Table 3 Hypotheses, tests and outcomes. H1–H3 were registered on 13 October 2026 before the benchmark, H4–H7 on 2 November 2026 before the corresponding experiments (Supplementary Sect. S2). Counts give the datasets (or dataset–noise combinations) in favour of LOOM; p is Holm-adjusted.

ID	Hypothesis	Metric	Result	Outcome
H1a	Clean NLL not worse than PreVal, hold-out TS, Platt, LogReg	NLL	vs PreVal: not significantly different, with 49 wins and 65 losses of 114 datasets, $p = 1.000$	pass
H1b	Clean ECE and Brier not worse than PreVal	ECE, Brier	ECE: LOOM worse on 86 of 114 datasets, $p < 0.001$	**fail**
H1c	Accuracy unchanged vs PreVal	accuracy	not significantly different, with 28 wins and 29 losses of 114 datasets, $p = 0.572$	pass
H2a	Faults: NLL lower than every closed-form / CV readout	NLL, 18 conditions	vs PreVal: LOOM better on 108/114, $p < 0.001$; all 7 readouts	pass
H2b	Faults: better detection of faulty inputs than PreVal	AUROC	LOOM better on 114/114, $p < 0.001$	pass
H3	Jackknife+: coverage 0.88–0.92, smaller sets than split	coverage, size	coverage 0.898; smaller on 103/114, $p < 0.001$	pass
H4	Real ECG noise: NLL lower than PreVal	NLL, 63 cells	LOOM better on 50/63, $p < 0.001$	pass
H5	Triage: more faulty recordings caught than PreVal	20 % budget	LOOM better on 38/38, $p < 0.001$	pass
H6	Triage: fewer automatic errors than PreVal	20 % budget	LOOM better on 35/38, $p < 0.001$	pass
H7	QUANT features: NLL under faults lower than PreVal	NLL, severity 3	LOOM better on 110/110, $p < 0.001$	pass

of PreVal and LOOM. QUANT (Dempster et al. 2024) and the Arsenal (Middlehurst et al. 2021) work on the raw series. Eight ablations remove or replace one component of LOOM at a time (Sect. 6.7). All methods receive the same data, seeds and fixed hyperparameter grids.

5.6 Hypotheses and tests

Table 3 lists the hypotheses with their metrics and outcomes. Datasets are the unit of analysis, with each metric averaged over resamples. We use two-sided Wilcoxon signed-rank tests (Wilcoxon 1945) with Holm's correction (Holm 1979), as recommended for comparisons across datasets (Benavoli et al. 2016; Demšar 2006), report how many datasets a result holds in, means with 95 % bootstrap confidence intervals (1,000 resamples), and Bayesian signed-rank tests (Benavoli et al. 2017) with a region of practical equivalence of ±0.01 nats for NLL and ±1 percentage point for rates. Numeric pass criteria for H1–H3 were written down after the results (deviation D18).

6 Results and analysis

6.1 Hypothesis tests

Table 3 summarises the outcomes and Fig. 2 the main comparison. Of the 10 prespecified hypotheses, nine passed and H1b failed.

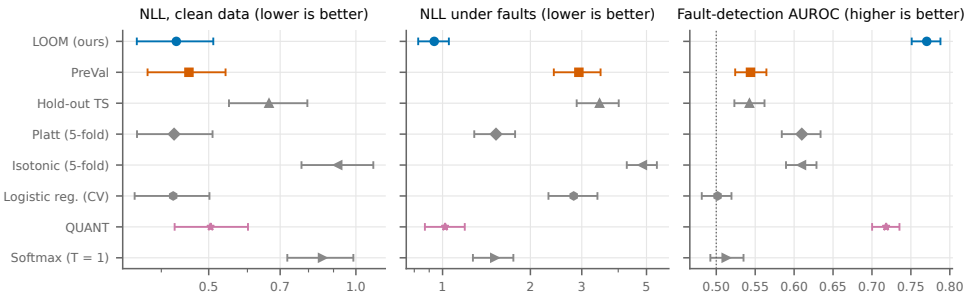


Fig. 2 LOOM is on a par with the best readouts on clean data and clearly better under simulated faults. Means over the 114 UCR datasets of the per-dataset mean over resamples, with 95 % bootstrap confidence intervals: NLL on clean data, NLL averaged over 18 simulated fault conditions (log scale), and AUROC for detecting faulty recordings. Ridge one-hot is omitted because its NLL is off scale.

Table 4 Clean test data, 114 datasets: median over datasets of the mean over resamples. W/D/L counts datasets on which LOOM is better/equal/worse in NLL; p is the Holm-adjusted Wilcoxon p-value for NLL.

Method	NLL	ECE	Brier	Acc.	W/D/L	p
LOOM (ours)	0.295	0.075	0.144	0.914	–	–
PreVal	0.360	0.040	0.134	0.914	49/0/65	$p = 1.000$
Hold-out TS	0.524	0.052	0.177	0.896	85/0/29	$p < 0.001$
Ridge + Platt (5-fold)	0.294	0.080	0.148	0.912	53/0/57	$p = 1.000$
Ridge + isotonic (5-fold)	0.670	0.044	0.157	0.904	96/0/14	$p < 0.001$
Logistic regression (CV)	0.314	0.049	0.151	0.901	53/0/56	$p = 1.000$
QUANT	0.386	0.116	0.210	0.895	77/0/37	$p < 0.001$
Ridge softmax ($T = 1$)	0.588	0.249	0.284	0.914	110/0/4	$p < 0.001$
Ridge one-hot	2.959	0.086	0.171	0.914	103/0/11	$p < 0.001$

6.2 Calibration on clean data (H1)

On clean data LOOM is roughly on a par with the best readouts but not better (Table 4). Its median NLL is 0.295; compared with PreVal it is not significantly different, with 49 wins and 65 losses of 114 datasets, $p = 1.000$, and it is also not significantly different from five-fold Platt scaling and logistic regression. A non-significant test does not mean equivalence: the Bayesian analysis gives probability 0.38 that LOOM is better, 0.00 that the two are practically equivalent and 0.62 that PreVal is better, and the per-dataset differences exceed resampling noise in favour of PreVal on 38 % of the datasets and in favour of LOOM on 13 % (Sect. 6.10). LOOM's calibration error is higher than PreVal's (median ECE 0.075 vs 0.040; LOOM worse on 86 of 114 datasets, $p < 0.001$), and so is its Brier score; H1b therefore fails. The cause is the target smoothing (Sect. 6.7). Accuracy is the same as PreVal's (median 0.914 vs 0.914).

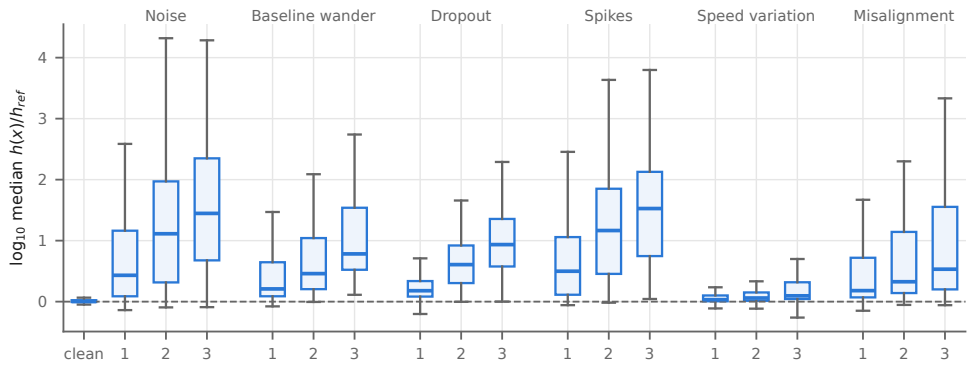


Fig. 3 Faults push recordings away from the training data, except speed variation. Distribution over datasets of the median leverage ratio $h(x)/h_{\text{ref}}$ ($\log_{10}$) for clean data and each simulated fault at severities 1–3.

Table 5 Simulated faults, 114 datasets: per-dataset mean over 18 conditions, then median over datasets. Statistics as in Table 4.

Method	NLL	ECE	Acc.	AUROC	W/D/L	p
LOOM (ours)	0.738	0.151	0.695	0.765	–	–
PreVal	1.728	0.185	0.692	0.563	108/0/6	$p < 0.001$
Hold-out TS	2.221	0.197	0.676	0.558	110/0/4	$p < 0.001$
Ridge + Platt (5-fold)	1.125	0.144	0.697	0.610	86/0/24	$p < 0.001$
Ridge + isotonic (5-fold)	4.358	0.153	0.682	0.592	108/0/2	$p < 0.001$
Logistic regression (CV)	1.878	0.219	0.660	0.516	104/0/5	$p < 0.001$
QUANT	0.732	0.170	0.694	0.709	44/0/70	$p = 0.291$
Ridge softmax ($T = 1$)	1.086	0.256	0.696	0.531	105/0/9	$p < 0.001$
Ridge one-hot	10.495	0.304	0.696	0.500	114/0/0	$p < 0.001$

6.3 Simulated sensor faults (H2)

The leverage reacts to faults (Fig. 3). For clean test series the median ratio $h(x)/h_{\text{ref}}$ is 1.01; at the highest severity it reaches 30.3 for noise, 29.5 for spikes, 8.6 for dropout, 6.2 for baseline wander and 3.5 for misalignment, but only 1.26 for speed variation. Averaged over the 18 conditions (Table 5), LOOM's median NLL is 0.738 against 1.728 for PreVal; LOOM is better on 108 of 114 datasets, $p < 0.001$ (Bayesian probability 1.00 that LOOM is better; mean NLL with 95 % CI 0.936 [0.825, 1.051] vs 2.932 [2.406, 3.480]), and it is also better than hold-out, Platt and isotonic calibration and logistic regression. Accuracy is unchanged (median 0.695 vs 0.692): the gain comes from lower confidence on the damaged recordings that are misclassified. The median AUROC for detecting faulty recordings is 0.765 for LOOM and 0.563 for PreVal. Figure 4 shows that the gap grows with severity for every fault type except speed variation.

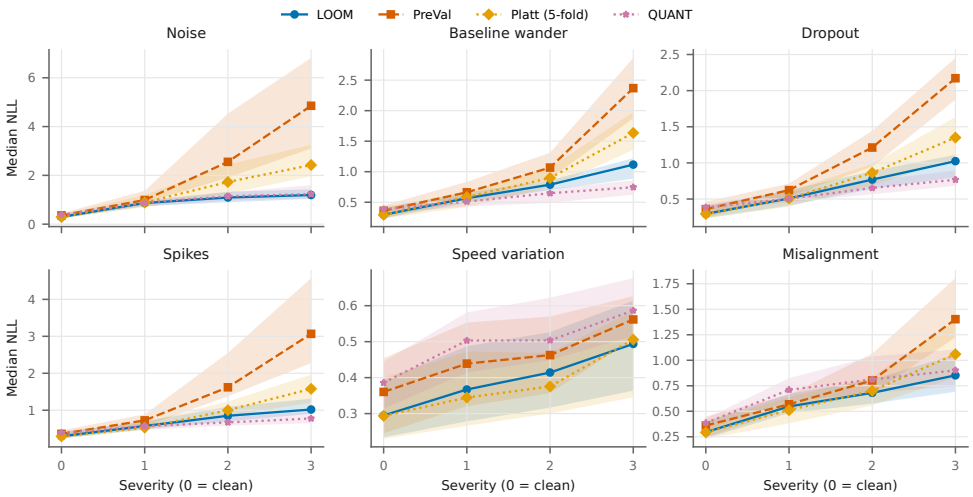


Fig. 4 LOOM's advantage over PreVal grows with fault severity, but QUANT is better for baseline wander, dropout and spikes at the highest severity. Median NLL over the 114 datasets against severity for each simulated fault, with 95 % bootstrap bands of the median.

6.4 Real-world validation: recorded ECG noise (H4)

Figure 5 and Table 6 show the results with real noise. All methods have a similar NLL on the clean ECG recordings (median 0.196 for LOOM, 0.175 for PreVal). With real noise added, PreVal's median NLL rises to 3.346 and LOOM's to 0.678. Over the 63 combinations of dataset, noise type and level, LOOM is better on 50 of 63 dataset–noise combinations, $p < 0.001$ compared with PreVal, confirming H4 (Bayesian probability 1.00 that LOOM is better), and not significantly different, with 28 wins and 35 losses of 63 dataset–noise combinations, $p = 0.748$ compared with QUANT. LOOM recognises noisy recordings best: its median detection AUROC is 0.973, against 0.899 for QUANT (LOOM better on 54 of 63 dataset–noise combinations, $p < 0.001$) and 0.591 for PreVal. In a stream with 30 % noisy recordings and a 20 % review budget, LOOM's queue contains 63 % of the noisy recordings, against 34 % for PreVal and 49 % for QUANT, while QUANT has the lowest automatic error rate (4.8 % vs 7.5 % for LOOM). With seven datasets, dataset-level tests have little power; the conclusions rest on the 63 combinations.

6.5 Case study: triage of faulty recordings (H5, H6)

In the exploratory study (12 datasets, 20 % review budget), LOOM's review queue contained a median of 45 % of the faulty recordings, against 25 % for PreVal and 38 % for QUANT, and caught more than PreVal on all twelve datasets ($p = 0.002$). Its automatic error rate was lower in the median (4.2 % vs 9.5 %) but not significantly ($p = 0.056$), and QUANT made the fewest automatic errors (1.6 %).

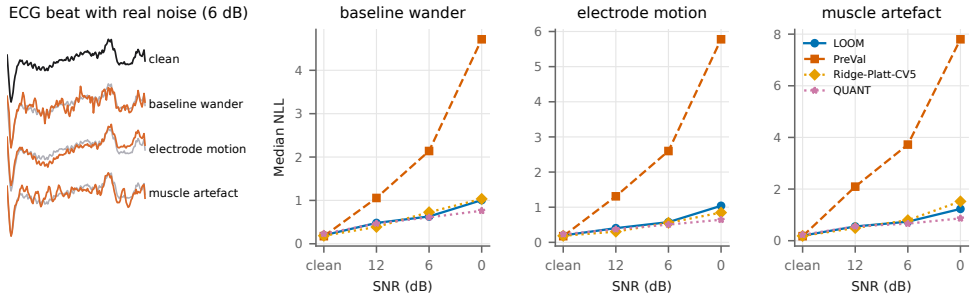


Fig. 5 Under real recorded ECG noise, PreVal's NLL explodes while LOOM's stays close to QUANT's. Left: an ECG5000 beat (grey) with each MIT-BIH NSTDB noise type at 6 dB (orange). Right: median NLL over the seven ECG datasets against signal-to-noise ratio for each noise type.

Table 6 Real recorded ECG noise on seven ECG datasets: median over datasets. Noisy NLL, detection AUROC and accuracy average nine combinations of noise type and SNR; "Caught" and "Auto err." refer to a stream with 30 % noisy recordings and a 20 % review budget.

Method	Clean NLL	Noisy NLL	AUROC	Acc.	Caught	Auto err.
LOOM (ours)	0.196	0.678	0.973	0.684	63 %	7.5 %
PreVal	0.175	3.346	0.591	0.673	34 %	9.2 %
Hold-out TS	0.194	6.097	0.623	0.675	37 %	10.2 %
Ridge + Platt (5-fold)	0.186	0.686	0.737	0.686	44 %	9.0 %
QUANT	0.231	0.585	0.899	0.700	49 %	4.8 %

The pre-registered confirmatory replication on 38 further datasets (Fig. 6, Table 7) confirmed both hypotheses under Holm's correction over all eight primary tests. LOOM's queue caught a median of 41 % of the faulty recordings, against 24 % for PreVal (LOOM better on 38 of 38 datasets, $p < 0.001$; H5) and 34 % for QUANT (LOOM better on 34 of 38 datasets, $p < 0.001$). Its automatic error rate was 10.1 %, against 14.6 % for PreVal (LOOM better on 35 of 38 datasets, $p < 0.001$; H6) and 13.2 % for QUANT (LOOM not significantly different, with 17 wins and 20 losses of 38 datasets, $p = 0.635$). Means with 95 % CIs are 43.4 % [39.4, 47.5] (LOOM) and 26.0 % [23.2, 29.2] (PreVal) for faults caught. The advantage over PreVal remains after merging datasets from the same source (LOOM better on 27 of 27 source-level units, $p < 0.001$). On fault-free streams the methods behave alike (automatic error 4.4 % vs 4.8 %), so LOOM does not change how a system behaves on good data. For an operator, this means that with one recording in five sent to review, about 41 % instead of 24 % of the damaged recordings reach a person.

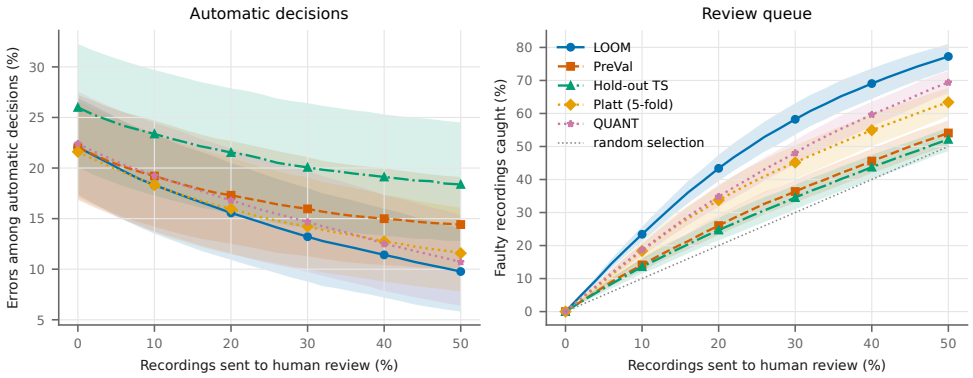


Fig. 6 In the pre-registered triage study, LOOM's review queue catches the most faulty recordings at every budget. 38 sensor datasets with 30 % faulty recordings: error rate of automatic decisions (left) and share of faulty recordings caught (right) against the review budget; means over datasets of the mean over resamples with 95 % bootstrap bands over datasets (bands include the differences in difficulty between datasets; paired comparisons are in the text).

Table 7 Triage with a 20 % review budget: median over datasets of the share of faulty recordings caught and of the error rate of automatic decisions, in the exploratory (12 datasets) and confirmatory (38 datasets) studies, and the automatic error on fault-free streams (confirmatory).

	Exploratory (12)		Confirmatory (38)		
Method	Caught	Auto err.	Caught	Auto err.	No faults
LOOM (ours)	45 %	4.2 %	41 %	10.1 %	4.4 %
PreVal	25 %	9.5 %	24 %	14.6 %	4.8 %
Hold-out TS	26 %	9.9 %	23 %	19.8 %	9.8 %
Ridge + Platt (5-fold)	27 %	8.7 %	28 %	15.0 %	4.6 %
QUANT	38 %	1.6 %	34 %	13.2 %	8.9 %

6.6 Comparative analysis: ensembles, raw-series classifiers and QUANT features (H7)

On the official split (Table 8) the comparison under faults uses the six severity-3 conditions for every method, because the Arsenal was only run there; this restriction was introduced after the results were first inspected and works against LOOM (deviation D10). On clean data a five-seed PreVal ensemble is slightly better than a single LOOM model (LOOM worse on 83 of 114 datasets, $p < 0.001$), while LOOM beats QUANT (LOOM better on 70 of 114 datasets, $p = 0.004$). Under the strongest faults LOOM beats the PreVal ensemble and the Arsenal, but QUANT is better (LOOM worse on 78 of 114 datasets, $p = 0.042$); averaged over all 18 conditions the two do not differ significantly (Table 5), but the Bayesian probability that QUANT is better is 0.86. At the highest severity QUANT has the lower median NLL for baseline wander, dropouts and spikes, and LOOM for noise, speed variation and misalignment. The Arsenal's

Table 8 Official split, strongest (severity-3) simulated faults: median over datasets. Statistics as in Table 4. The Arsenal has no result on Fungi (undefined probabilities with one training series per class).

Method	NLL	Brier	Acc.	AUROC	W/D/L	p
LOOM (ours)	1.020	0.526	0.604	0.845	–	–
LOOM, 5-seed ens.	1.003	0.519	0.614	0.847	38/0/76	$p < 0.001$
PreVal	2.260	0.673	0.613	0.531	105/0/9	$p < 0.001$
PreVal, 5-seed ens.	2.040	0.636	0.616	0.584	104/0/10	$p < 0.001$
QUANT	0.931	0.489	0.621	0.760	36/0/78	$p = 0.042$
Arsenal	7.024	0.672	0.600	0.676	110/0/3	$p < 0.001$

high NLL comes mainly from vote-based probabilities that give exactly zero to the true class; in the bounded Brier score the clean gap is small (0.173 vs 0.163).

To see whether QUANT's robustness comes from its features or its trees, we fitted the readouts on QUANT's features (median 2,957 features, 110 datasets). LOOM again rescued the ridge readout under faults: PreVal's median NLL was 17.7 and LOOM's 1.242 (LOOM better on 110 of 110 datasets, $p < 0.001$), confirming H7. LOOM on QUANT features also detected faults far better than the QUANT classifier (AUROC 0.950 vs 0.767; LOOM better on 93 of 110 datasets, $p < 0.001$). But the linear readout lost much more accuracy than QUANT's trees (0.354 vs 0.621), so QUANT kept the lower NLL (LOOM worse on 97 of 110 datasets, $p < 0.001$). QUANT's robustness therefore comes mainly from its trees, whose outputs stay within the range seen in training, while LOOM's strength is recognising unusual inputs; the two are complementary.

6.7 Ablation study

Figure 7 compares LOOM with eight variants. Removing the leverage scaling while keeping smoothing (PreVal-smooth) is much worse under faults (median NLL 1.172 vs 0.738; LOOM better on 94 of 114 datasets, $p < 0.001$), while on clean data LOOM is worse on 75 of 114 datasets, $p = 0.038$. Calibrating on in-sample instead of LOO pairs gives a median clean NLL of 0.978, and an 80/20 split 0.441, so the exact LOO pairs matter. A finite moderation strength ($c = 1$) gives a slightly lower clean NLL (LOOM worse on 89 of 114 datasets, $p < 0.001$) but detects faults worse (AUROC 0.729 vs 0.765; LOOM higher on 111 of 114 datasets). Removing the smoothing brings the clean ECE to PreVal's level (0.041 vs 0.040) but is worse under faults (LOOM better on 71 of 114 datasets, $p = 0.003$). A nearest-neighbour distance in place of leverage gives similar NLL but worse detection (AUROC 0.725).

6.8 Prediction sets and validity (H3)

On clean data LOOM's closed-form jackknife+ sets with the LAC score cover 89.8 % of test series at target 0.9 with 1.55 labels per set, against 92.8 % and 2.49 for split conformal prediction (set size smaller on 103 of 114 datasets, $p < 0.001$), so H3 passes (Fig. 8a, Table S3). Coverage at targets 0.95 and 0.8 is 95.1 % and 79.6 %. Under faults coverage drops because damaged recordings are not exchangeable with training

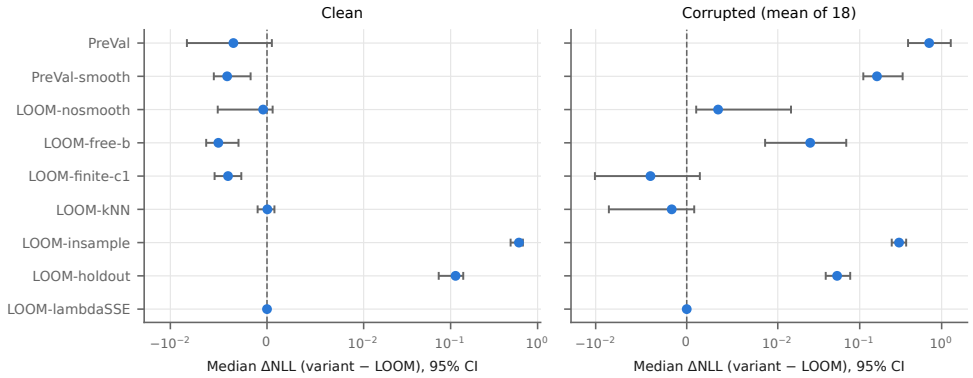


Fig. 7 The leverage scaling and the exact LOO pairs are what make LOOM work under faults. Median paired NLL difference (variant − LOOM) over the 114 datasets with bootstrap 95 % confidence intervals, clean data (left) and simulated faults (right); positive values mean that the variant is worse.

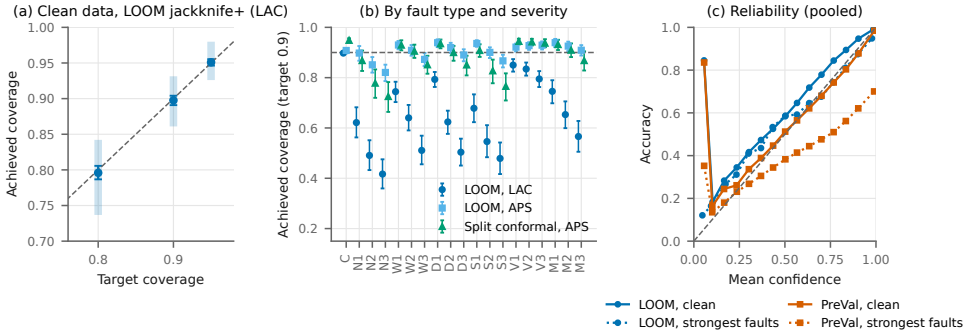


Fig. 8 Jackknife+ coverage matches its target on average on clean data; under faults only APS sets keep coverage. (a) Target vs achieved coverage of LOOM's jackknife+ sets (LAC), mean over datasets with 95 % bootstrap CI and 10th–90th percentile of datasets (band). (b) Achieved coverage at target 0.9 by condition (C clean; N noise, W baseline wander, D dropout, S spikes, V speed variation, M misalignment; 1–3 severity). (c) Reliability pooled over all datasets and resamples; bins with fewer than 200 predictions are not drawn.

data (Fig. 8b); with LAC, LOOM's flat probabilities produce empty sets for 25.0 % of damaged recordings, whereas APS keeps coverage at 90.5 %. The reliability diagrams (Fig. 8c) show that LOOM stays close to the diagonal under strong faults while PreVal becomes overconfident.

6.9 Qualitative analysis and failure cases

Figure 9 shows four recordings of the ECG5000 real-noise stream, chosen by a fixed rule. Two show the intended behaviour; two are failures. A recording with muscle

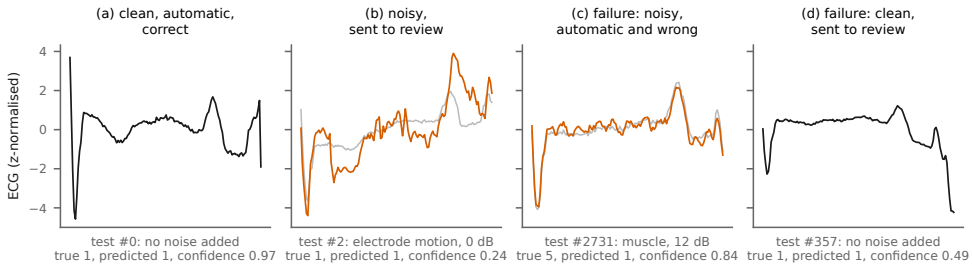


Fig. 9 LOOM flags most noisy recordings but misses some and sends some clean ones to review. Real-noise stream of ECG5000 (official split, LOOM, 20 % review budget). Rule: for each outcome, the first test recording in file order with that outcome. Grey: clean recording; orange: with NSTDB noise.

artefact at 12 dB is misclassified with high confidence (panel c), because the noise barely changes its leverage; and a clean but unusual beat is sent to review (panel d). Across datasets, LOOM's gain under faults grows with the leverage increase the faults cause (Spearman $\rho = 0.61$, $p < 0.001$), and the share of datasets on which LOOM beats PreVal grows with severity for every fault type except speed variation. At the highest severity it is 95 % for noise but only 53 % for speed variation. LOOM loses most where faulty recordings remain correctly classified (PigCVP, ECG5000 and InsectEPGRegularTrain; $\rho = -0.24$ with accuracy under faults, $p = 0.010$), and on clean data where there are few training series per class (largest losses on PigArtPressure, Symbols, ShapesAll and PigAirwayPressure, with two to 10 series per class), because smoothing then makes the model underconfident.

6.10 Generality, sensitivity and sanity checks

With MultiRocket features LOOM is better on 105 of 110 datasets, $p < 0.001$ under faults and worse on 72 of 110 datasets, $p = 0.008$ on clean data; with Hydra features better on 106 of 110 datasets, $p < 0.001$ and not significantly different, with 43 wins and 67 losses of 110 datasets, $p = 0.147$. On the 100 datasets never used during development LOOM is better on 94 of 100 datasets, $p < 0.001$ under faults; without datasets with exact duplicates better on 92 of 98 datasets, $p < 0.001$; without exact or near duplicates (90 datasets) better on 84 of 90 datasets, $p < 0.001$; and after merging datasets from the same source (70 units) better on 68 of 70 source-level units, $p < 0.001$. With 25 % and 50 % of the training data, LOOM's median NLL under severity-3 faults is 1.058 and 1.049, against 2.155 and 2.543 for PreVal (111 datasets); LOOM keeps its advantage under faults with smaller training sets. Under a real change of sensor (training on one accelerometer axis of UWaveGestureLibrary and testing on another), accuracy drops from 0.810 to 0.323 but leverage barely moves (median ratio 1.13–1.29), and LOOM's advantage is small (mean NLL 2.469 vs 2.550, lower on six of six axis pairs).

Sanity checks. (1) Assumption A2: held-out and clean test leverages have the same median scale, but a Kolmogorov–Smirnov test rejects equal distributions on 19 of

110 datasets (Holm-adjusted), so the assumption holds only approximately. (2) The spread of per-dataset jackknife+ coverage (standard deviation 0.035) is larger than binomial sampling alone would give (0.010, 97.5 % quantile 0.013); 42 % of datasets lie outside the binomial 95 % range, so coverage holds on average but not for every dataset. (3) Differences between LOOM and PreVal exceed two resampling standard errors on 51 % of the datasets on clean data and on 78 % under faults, in favour of LOOM on 76 % of the datasets.

6.11 Computational complexity and cost

LOOM costs the same to fit as PreVal (median 0.14 s vs 0.14 s on top of the shared MiniRocket transform of 0.17 s; Supplementary Table S4). From raw data to a fitted model, LOOM takes about 0.31 s, against 1.30 s for QUANT, 2.25 s for a five-seed ensemble and 4.47 s for the Arsenal.

6.12 Software tool

We release LOOM as a Python package with a command-line tool, `loom-triage`: `fit` trains a model on a CSV of labelled recordings, `run` writes a review queue with prediction, confidence, unusualness (h/h_{ref}), decision and optional prediction set, plus an HTML report, and `evaluate` scores labelled data. The tool has its own unit tests and was validated end to end on plain CSV files outside the experimental pipeline (Supplementary Sect. S5).

6.13 Discussions

LOOM uses a second quantity that the LOO algebra provides for free, the held-out leverage, to put each prediction on the scale of its own uncertainty. For an operator, this means that a LOOM model that is unsure about a recording is much more likely to be looking at a damaged one, and that a fixed review budget catches about 41 % rather than 24 % of the damaged recordings in our confirmatory study. QUANT classifies damaged recordings more accurately; it made the fewest automatic errors in the exploratory study and with real ECG noise, while in the confirmatory study LOOM's median automatic error was lower but not significantly. A tree ensemble for the label combined with LOOM's leverage for the review decision is a promising design. We would use LOOM when training sets are small, recordings may be damaged, and a single fast model is preferred; where test data are known to be clean, PreVal gives a lower clean calibration error.

The main limitations are:

1. Apart from the real ECG noise, all faults are simulated, and real noise was tested on ECG only. The only real change of sensor we studied barely moves leverage, and LOOM gains little there.
2. Leverage does not respond to changes that keep the features within the training range (speed variation, another sensor axis, label noise, changing class frequencies).
3. On clean data LOOM is not better than PreVal, and target smoothing raises its calibration error on datasets with few training series per class.

4. QUANT has a lower likelihood loss than LOOM under the strongest faults.
5. The jackknife+ guarantee holds only approximately (Assumption A4), and per-dataset coverage varies more than sampling noise allows.
6. For compute reasons, ensembles, other extractors and the Arsenal were evaluated on the official split only, logistic regression was skipped for four large datasets, and deep networks were not run.
7. The exploratory triage study and several analyses were designed after the main results were known; they are labelled as post hoc (Supplementary Sect. S2).

7 Conclusion and future scope

We showed that a closed-form change to the readout of fast random-kernel classifiers makes their confidence drop for recordings far from the training data, without held-out data, refitting or ensembles. LOOM lowered the likelihood loss under simulated faults and real ECG noise compared with the strongest previous readout, and in a pre-registered triage study its review queue caught 41 % instead of 24 % of the faulty recordings. We did not show an advantage on clean data or over the tree-based QUANT classifier under the strongest faults, and leverage does not detect changes that keep the features in range. Next steps are a study with expert-labelled real faulty recordings, multivariate signals, combining QUANT's trees with LOOM's leverage, and adaptive target smoothing for problems with many classes and few examples.

Supplementary information. Supplementary Material: proofs, deviations log, leakage audit, fault definitions, software validation and additional tables and figures.

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- Ethics approval and consent to participate: not applicable (public, de-identified benchmark data).
- Data availability: UCR archive (https://www.timeseriesclassification.com) and MIT-BIH Noise Stress Test Database (https://physionet.org/content/nstdb/1.0.0/).
- Code availability: [public repository link to be inserted by the authors]; all results, tables, figures and the number macros of this paper are regenerated by one script.
- Author contribution: [to be completed by the authors]

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